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Copper-Catalyzed Tandem Multi-Component Approach to 1,3-Oxazines at Room Temperature by Cross-Dehydrogenative Coupling Using Methanol as C1 Feedstock

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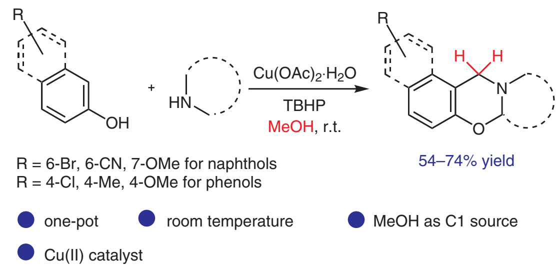

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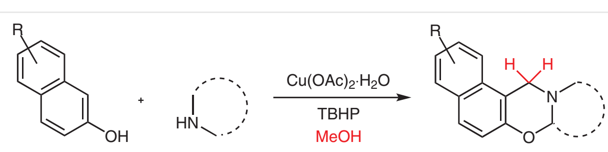
Abstract A copper(II)-catalyzed multi-component one-pot approach for the synthesis of 1,3-oxazines at room temperature is reported here. Methanol is used as the solvent as well as the carbon source. The methylene carbon of the oxazine product comes from methanol via formaldehyde. *tert*-Butyl hydroperoxide is used as the oxidant. The reaction uses an environmentally benign metal catalyst and oxidant. No inert atmosphere or precaution is required for the reaction. Most importantly, the reaction avoids the use of carcinogenic formaldehyde.

Key words methanol, 1,3-oxazine, copper(II) catalyst, multi-component reaction, C–H functionalization

The development of sustainable processes for organic synthesis which exploit inexpensive and easily available feedstocks and reagents has recently become a centre of attraction in chemical research.[1] For this purpose, methanol has appeared as a renewable resource with high potential which is used to produce plentiful daily-life commodities and industrial chemicals.[2] It is a very simple aliphatic, cheap and biodegradable alcohol, which is used as an alternative fuel for internal combustion.[3] Usually, methylation of organic molecules is achieved by using toxic methyl compounds such as methyl iodide, dimethyl sulfate and diazomethane,[4] or by using carcinogenic formaldehyde.[5] However, methanol can be used as a C1 source through the in situ generation of formaldehyde by dehydrogenative activation of methanol. The formaldehyde thus generated can be transformed to methyl, methylene and hydroxylmethyl functional groups.[6] Researchers have even used methanol as a carbonyl source to synthesize urea derivatives.[7] Usually Ru, Ir and Rh complexes are employed to catalyze these reactions.[6–8] However, rarely a Cu catalyst is used.[9]

1,3-Oxazine derivatives have shown anti-HIV, antitumor, antibacterial, antituberculosis, fungicidal and many other activities.[10] In addition, specifically 1,3-naphthoxazine derivatives are highly useful for the treatment of Parkinson's disease and as potent nonsteroidal progesterone receptor agonists.[11] Many synthetic routes have been developed for their preparation;[12] however, because the cross-dehydrogenative coupling (CDC) technique offers many advantages such as atom-economical, directive and environmentally benign properties and non-requirement of prefunctionalized substrates,[13] synthetic chemists also synthesized these compounds by using this strategy.[14]

Moreover, multi-component reactions are one of the most powerful techniques in synthetic organic chemistry for the synthesis of highly functionalized organic molecules.[15] They have gained much attention because of their advantages such as atom and step economy, simpler procedures and as an energy-saving synthetic tactic to generate new bonds in a single step.[16] By using these techniques, we herein disclose a tandem three-component route to 1,3-oxazines in methanol as the solvent as well as C1 source by CDC (Scheme 1). To the best of our knowledge, this is the first report of synthesizing 1,3-oxazines with use of methanol as the C1 source.



Scheme 1 Synthetic route to 1,3-oxazines

We chose the combination of 2-naphthol (**1a**) and tetrahydroisoquinoline (**2a**) in MeOH using different catalysts/oxidants to synthesize 1,3-oxazine **3a**. By using Fe(III), Ru(III) and I_2 catalysts in the presence of TBHP as the oxidant no reaction was observed. Copper(II) acetate monohydrate (10 mol%) in the presence of TBHP (2 equiv) gave **3a** in 56% yield (Table 1, entry 4). However, increasing the amount of solvent from 2 to 5 mL increased the yield and we isolated product **3a** in 68% yield; this were the optimal conditions for this reaction (Table 1, entry 11). The reason may be that in higher dilution the unidentified little impurities which otherwise formed were not observed. However, further dilution lowered the product yield (Table 1, entry 12). Additional changes of the oxidant to H_2O_2 or oxone gave lower yield or no yield (Table 1, entries 5–6). Other copper catalysts such as CuBr, $CuSO_4{\cdot}5H_2O$ and copper triflate offered lower yields (Table 1, entries 8–10). An increase of the reaction time did not improve the yield (Table 1, compare entry 11 and entry 13). At higher temperature the reaction was unsuccessful to form the product (Table 1, entry 7). Increase of catalyst/oxidant loading did not affect the yield (Table 1, entries 16 and 14), but reduction of their amount reduced the yield to a little extent (Table 1, entries 17 and 15).

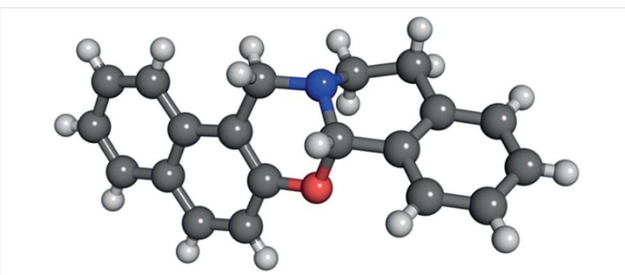


Figure 1 Single-crystal X-ray structure of **3a**

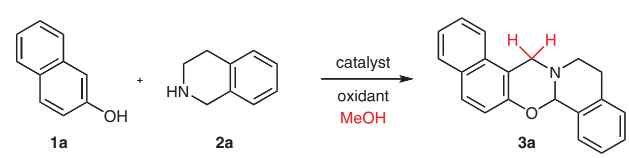
After establishing the optimal conditions, we subsequently investigated the substrate scope for the synthesis of **3**, and the products were characterized by NMR and mass spectroscopy as well as single-crystal X-ray crystallography (see Figure 1 and Figure 2).[17,18] We screened different amines such as pyrrolidine, piperidine and obtained moderate yields. Open-chain amines did not respond to this reaction. We also screened different naphthols and obtained optimistic results. Moreover, we used phenolic substrates for this reaction; however, they gave relatively low yields. Other alcohols such as ethyl, propyl or benzyl alcohol did not give positive results.

To gain insights in the mechanism for this reaction, we performed control experiments. A representative reaction was carried out in complete absence of air/O_2 (in N_2 atmosphere), which did not affect the yield. This ensures that molecular O_2 is not the oxidant in this reaction. We also performed the reaction in the presence of radical scavengers such as BHT and TEMPO (using 1.5 equiv of each), under the optimized conditions, which provided very low yield of **3a** (12% of **3a** was obtained by using BHT and 8% of **3a** by using TEMPO). These experiments indicate that the reaction proceeds through a free-radical route. Moreover, when we replaced MeOH with CD_3OD, 49% of deuterated product **3a'** was formed, which was confirmed through NMR and mass spectroscopy analysis. This suggests that the methylene carbon of the product absolutely comes from MeOH (Scheme 2).

To check the possibility of a [4+2] cycloaddition mechanism for this reaction, we performed a cross reaction between **2c'** and **2e'** (synthesized by a reported procedure[14a]) under the optimized conditions. However, we did not notice any cross product from the reaction and obtained only two normal CDC products, namely **3c** and **3e** (Scheme 3). Therefore, we believe that this method for the synthesis of 1,3-oxazines does not proceed through [4+2] cycloaddition of imine and *o*-quinone methide unlike the mechanism reported by Maycock et al.[14a]

Based on the above experiments, a tentative mechanism is proposed for the reaction (Scheme 4). First, TBHP in the presence of Cu(II) gives *tert*-butyl peroxy radical (**X**) and

Table 1 Optimization of the Reaction Conditions[a]



Entry	Catalyst (mol%)	Oxidant (equiv)	MeOH (mL)	Time (h)	Yield (%)
1	$FeCl_3{\cdot}6H_2O$ (10)	TBHP (2)	2	15	–
2	$RuCl_3{\cdot}3H_2O$ (10)	TBHP (2)	2	15	–
3	I_2 (10)	TBHP (2)	2	15	–
4	$Cu(OAc)_2{\cdot}H_2O$ (10)	TBHP (2)	2	15	56
5	$Cu(OAc)_2{\cdot}H_2O$ (10)	H_2O_2 (2)	2	15	34
6	$Cu(OAc)_2{\cdot}H_2O$ (10)	Oxone (2)	2	15	–
7	$Cu(OAc)_2{\cdot}H_2O$ (10)	TBHP (2)	2	5[b]	–
8	CuBr (10)	TBHP (2)	2	15	38
9	$CuSO_4{\cdot}5H_2O$ (10)	TBHP (2)	2	15	35
10	$Cu(OTf)_2$ (10)	TBHP (2)	2	15	14
11	$Cu(OAc)_2{\cdot}H_2O$ (10)	TBHP (2)	5	15	68
12	$Cu(OAc)_2{\cdot}H_2O$ (10)	TBHP (2)	7	15	62
13	$Cu(OAc)_2{\cdot}H_2O$ (10)	TBHP (2)	5	20	67
14	$Cu(OAc)_2{\cdot}H_2O$ (10)	TBHP (2.5)	5	15	68
15	$Cu(OAc)_2{\cdot}H_2O$ (10)	TBHP (1.5)	5	15	55
16	$Cu(OAc)_2{\cdot}H_2O$ (15)	TBHP (2)	5	15	66
17	$Cu(OAc)_2{\cdot}H_2O$ (5)	TBHP (2)	5	15	58

[a] Unless otherwise mentioned, all reactions were performed by using **1a** (1 mmol, 144 mg) and **2a** (1 mmol, 133 mg) in MeOH at room temperature. Products were purified by column chromatography and yields are given for the isolated products.
[b] Reaction was carried out under reflux.

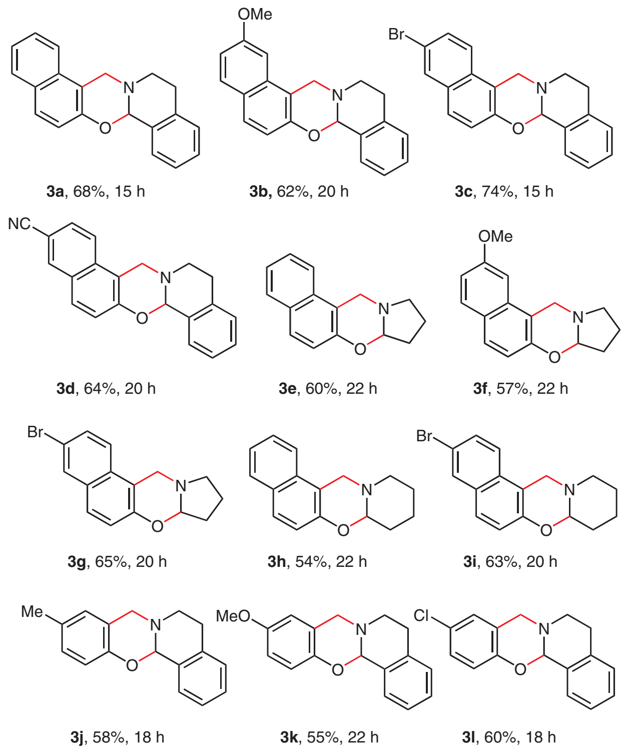
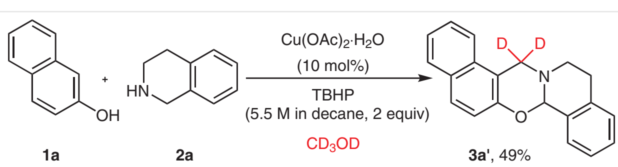
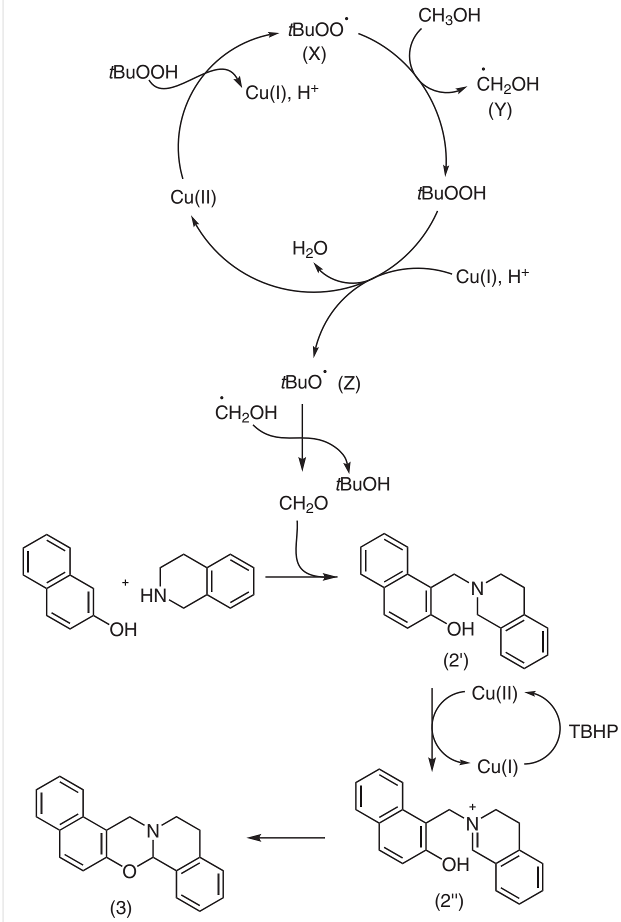


Figure 2 Substrate scope for the synthesis of compound **3**. Reaction conditions: **1** (1.0 mmol), **2** (1.0 mmol), $Cu(OAc)_2{\cdot}H_2O$ (10 mol%, 20 mg), TBHP (70% in H_2O, 2 equiv, 257 mg) in MeOH (5 mL) at room temperature. Products were purified by column chromatography using silica gel (100–200 mesh) and yields are given for the isolated products.

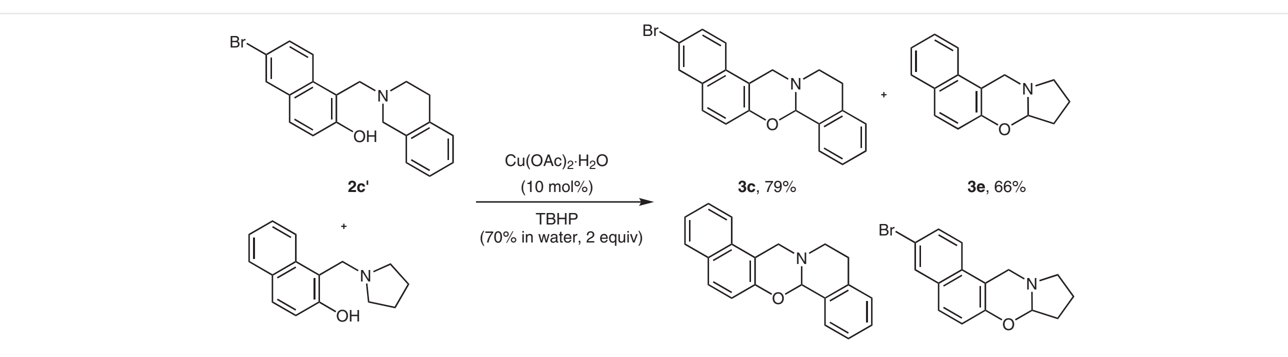


Scheme 2 Deuterium labeling for mechanistic investigation. This particular reaction was carried out by using TBHP in decane, because otherwise, in the presence of water, partial deuterium exchange occurred.



Scheme 4 Tentative mechanism for the reaction

Cu(II) is converted into Cu(I). Radical **X** then reacts with one molecule of MeOH generating hydroxymethyl radical (**Y**). Subsequently, TBHP in the presence of Cu(I) gives *tert*-butoxy radical (**Z**), which in the presence of a hydroxymethyl radical gives formaldehyde. Next, the 3-component Mannich reaction of naphthol, amine and formaldehyde gives α-alkyl amino naphthol **2'**, which is then converted into the



Scheme 3 Cross experiment for mechanistic proof

corresponding iminium ion **2''** in the presence of Cu(II) and finally cyclized to the desired product **3**.

In conclusion, we have successfully developed a 3-component route for the synthesis of 1,3-oxazines using methanol as C1 feedstock. The methylene carbon of the product originates from methanol. The reaction is carried out at room temperature and utilizes an environmentally benign metal catalyst. On the basis of our control experiments, a tentative free-radical mechanism is also proposed.

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Supporting Information

Supporting information for this article is available online at https://doi.org/10.1055/s-0036-1591775.

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(17) Crystallographic data for compound **3a** has been deposited with CCDC 1812944 and can be obtained free of charge from the Cambridge Crystallographic Data Centre, www.ccdc.cam.ac.uk/conts/retrieving.html.

(18) **Representative Procedure for the Synthesis of 3a**
2-Naphthol (**1a**, 1 mmol, 144 mg), tetrahydroisoquinoline (**2a**, 1 mmol, 133 mg), and methanol (5 mL) were combined in a round-bottom flask. $Cu(OAc)_2 \cdot H_2O$ (10 mol%, 20 mg) and TBHP (70% in H_2O, 2 mmol, 257 mg) were added and the reaction mixture was stirred at r.t. for 15 h. Progress of the reaction was monitored by TLC. Removal of the solvent under vacuum and purification by column chromatography (100–200 mesh silica gel, hexane/ethyl acetate) afforded product **3a** as a white solid. Yield: 68% (195 mg). ^{1}H NMR (300 MHz, $CDCl_3$): δ = 7.79 (d, *J* = 7.9 Hz, 1 H), 7.69–7.64 (m, 2 H), 7.53–7.45 (m, 2 H), 7.40–7.30 (m, 3 H), 7.23 (d, *J* = 7.9 Hz, 1 H), 7.07 (d, *J* = 8.7 Hz, 1 H), 5.80 (s, 1 H), 4.81 (d, *J* = 16.3 Hz, 1 H), 4.30 (d, *J* = 16.6 Hz, 1 H), 3.49–3.41 (m, 1 H), 3.17–3.12 (m, 1 H), 2.97–2.92 (m, 2 H). ^{13}C NMR (125 MHz, $CDCl_3$): δ = 151.7, 134.9, 133.0, 131.5, 129.0, 128.9, 128.8, 128.5, 128.2, 126.5, 126.3, 123.5, 121.2, 118.8, 111.1, 86.9, 51.3, 45.2, 29.2. HRMS (ESI): *m*/*z* calcd for $C_{20}H_{17}NO$ $[M+H]^+$: 288.1388; found: 288.1389.